

Haroun Guessous

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Skills

- C# | .NET | Java | JavaScript | TypeScript | C++ | C | CosmosDB | MSSQL | Node | React | jQuery | NoSQL | Git
- Azure | Cloud Computing | CI/CD | XUnit | Jest | Cucumber | Nightwatch | Unit Testing | Lambda | OOP | Unity 2D
- Microservices | Distributed Systems | Frontend | Backend | Full-Stack | English, French, Arabic – *All professional proficiency or above*

Education

Bachelor of Engineering

McGill University

- Major in Software Engineering, minor in Finance.
- Recipient of the Rubin Gruber Scholarship, the McGill merit-based Entrance Scholarship, and the MES Scholarship for Academic Outstanding, Dean's List 2020 and 2021

Experience

Software Engineer

CDPO

01/2024 - 04/2025

- Optimized an internal scheduling tool using C# and Angular, achieving a 40% performance boost, integrating NLP for conversational task management, and implementing metrics tracking to increase productivity by 25%.
- Developed a Python-based asset risk simulation tool with a React frontend and Flask backend, leveraging Snowflake for data processing, and improving prediction accuracy by 18%.
- Designed and implemented RESTful APIs using Flask to unify data exchange between internal systems, ensuring secure access and reducing manual reporting by 15 hours/week.
- Developed a comprehensive TypeScript package for optimized data processing and visualization, streamlining workflows and enhancing data display capabilities, resulting in improved system efficiency and an annual cost savings of approximately \$13,000.

Software Engineering Lead

McGill Formula Electric

09/2021 - 10/2024

- Developed the Anti-lock Braking System (ABS) for an electric formula car using C and C++, resulting in an 18% improvement in response time compared to the existing system.
- Integrated **Arduino** and assembly to enhance the software's electrical components, **reducing system response time by an additional 14%**.
- Utilized **C++** and **OpenCV** to build a computer vision software capable of measuring driver characteristics with 99.3% accuracy, aiding in the design of an optimal chassis.
- Created a machine learning algorithm in **TensorFlow** to predict tire wear patterns, improving predictive maintenance accuracy by 30%.
- Developed a real-time telemetry system using **Python**, enabling remote diagnostics and performance monitoring, which increased data processing speed by 25%.

Software Engineering Intern

Otera Capital

05/2023 - 01/2024

- Used Python's **Flask**, **Javascript**, **React.js** and numerous python analytical modules to create a simulation tool based on **Monte-Carlo Simulation** model to accurately predict the evolution of the company's portfolio, Which reduced the process of getting portfolio predictions overall time consumption by 500%.
- Implemented Docker and AWS for app deployment, integrated Amazon's Airflow and PostgreSQL to automate cashflow reports and currency rate updates for the simulation tool, resulting in a 40% reduction in report generation time.

Full Stack Software Developer intern

OneDesk

a 05/2022 - 09/2022

- Leveraged **Java**, **HTML**, and **Angular** to redesign and revamp the website and user portals to meet 508 Compliance standards, significantly enhancing accessibility and usability for individuals with disabilities. This allowed for seamless navigation and interaction, improving user satisfaction and engagement.
- **Reported directly to CPO:** Collaborated with **Spring** and **Hibernate** to introduce new features and optimize existing functionalities within the application, resulting in a more resource-efficient system. These improvements led to an 11% **reduction in page loading time** while simultaneously reducing memory usage by 9%, boosting overall app performance.
- Developed workflow automation systems using **C++** and **Java**, enabling users to configure custom triggers and automate app responses tailored to their preferences. This innovation resulted in an 87% reduction in the average time required to complete repetitive actions, driving productivity gains.

Projects

- [Single player First Person Shooter in C++](#) : C++ for core logic and interface, **OpenGL** for rendering, custom Physics Engine
- [5LeagueFantasy](#) : Fantasy Soccer web-app, Backend in **Java** and **spring boot**, frontend in **React.js**
- [x86 Linux debugger](#) : Miniature x86 linux debugger built using **C++**
- [FPGA based Transformer](#) : Using **Python** and VHDL to build **NanoGPT** on an **FPGA** under the supervision of Prof Brett H. Meyer